Lab 2: Transformers for In-Context Regression

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The data

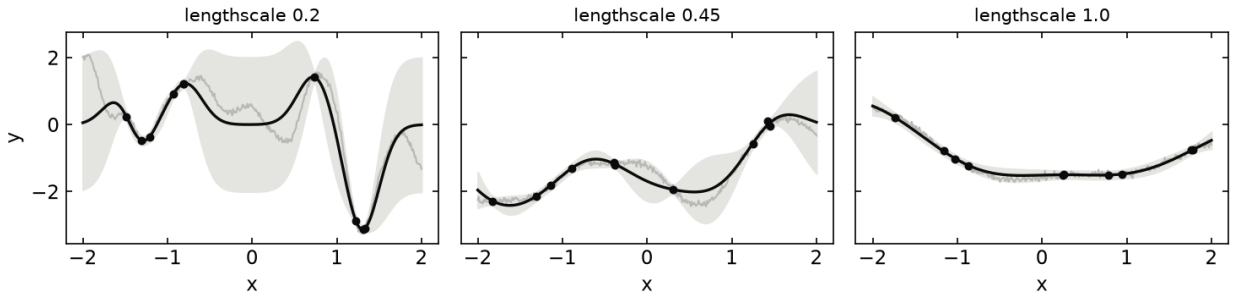


Figure 1: Example functions drawn from the Gaussian process at three lengthscales, with 10 noisy context points (black), the true function (light gray), and the exact GP posterior mean $\pm 2\sigma$.

Transformer

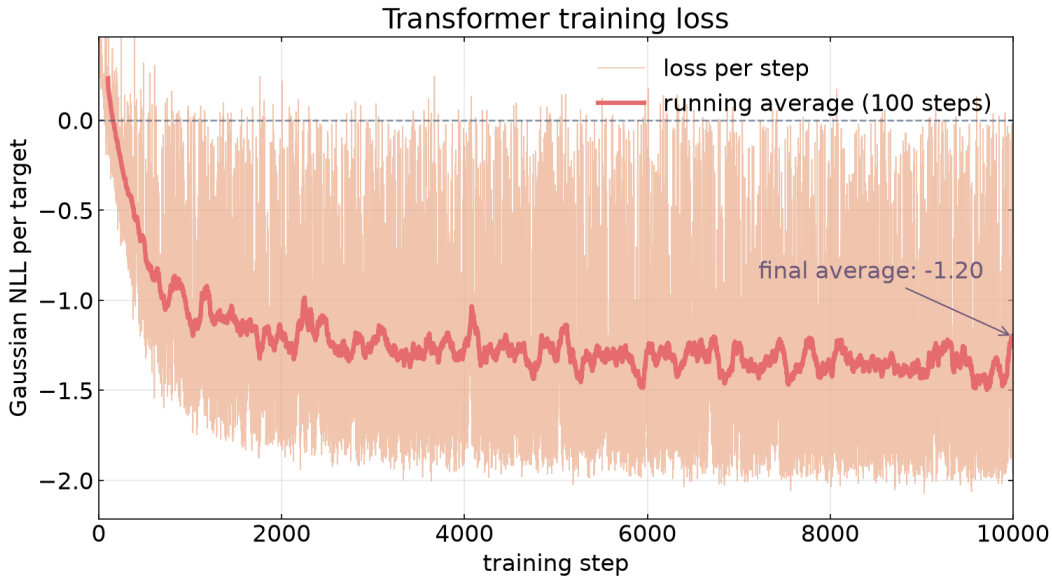


Figure 2: Transformer training loss (Gaussian NLL per target) per step, with a 100-step running average.

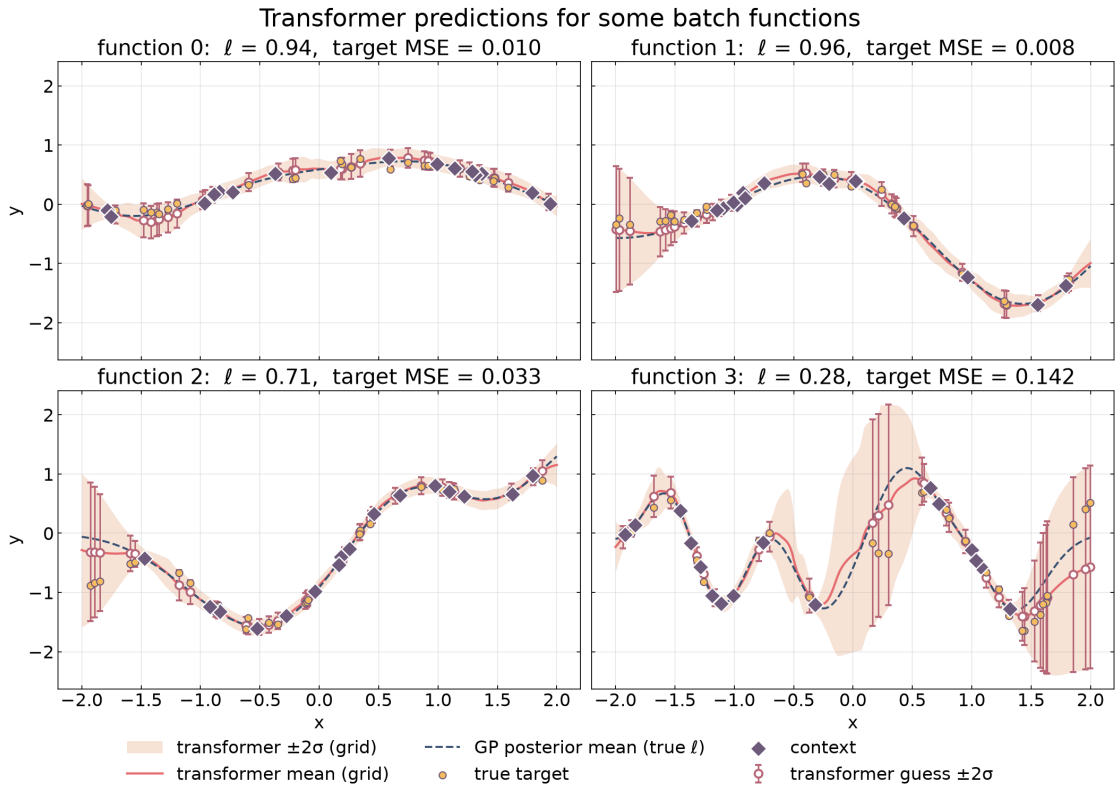


Figure 3: Transformer predictions for four functions: guesses at the target points with $\pm 2\sigma$ error bars, the prediction on a dense grid, the true targets, the context, and the exact GP posterior mean.

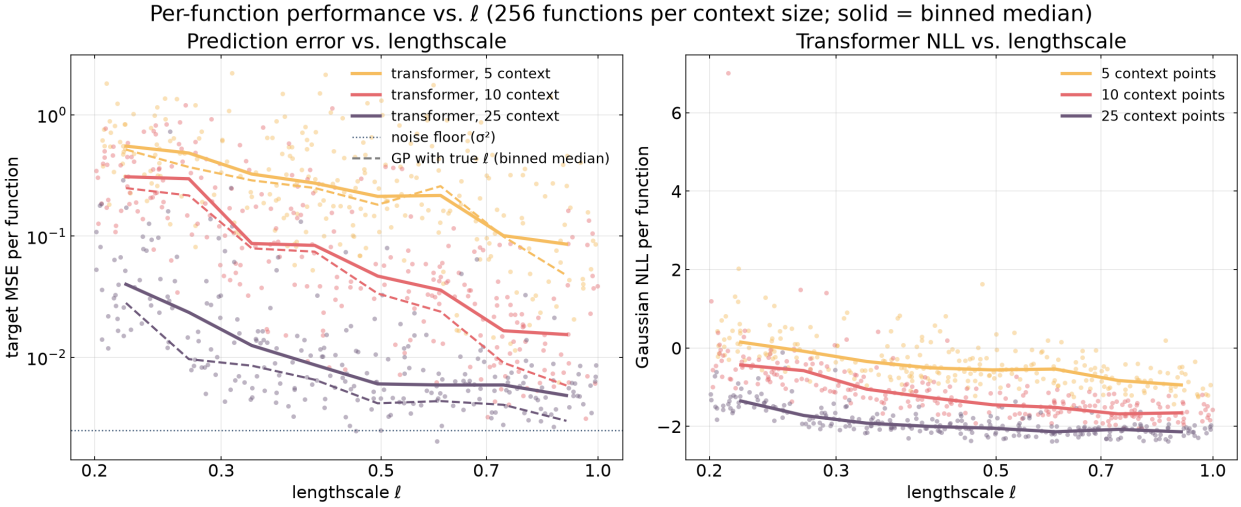


Figure 4: Per-function transformer error (left, with the GP for comparison) and NLL (right) against the lengthscale ℓ, for 5, 10 and 25 context points.

MLP comparison

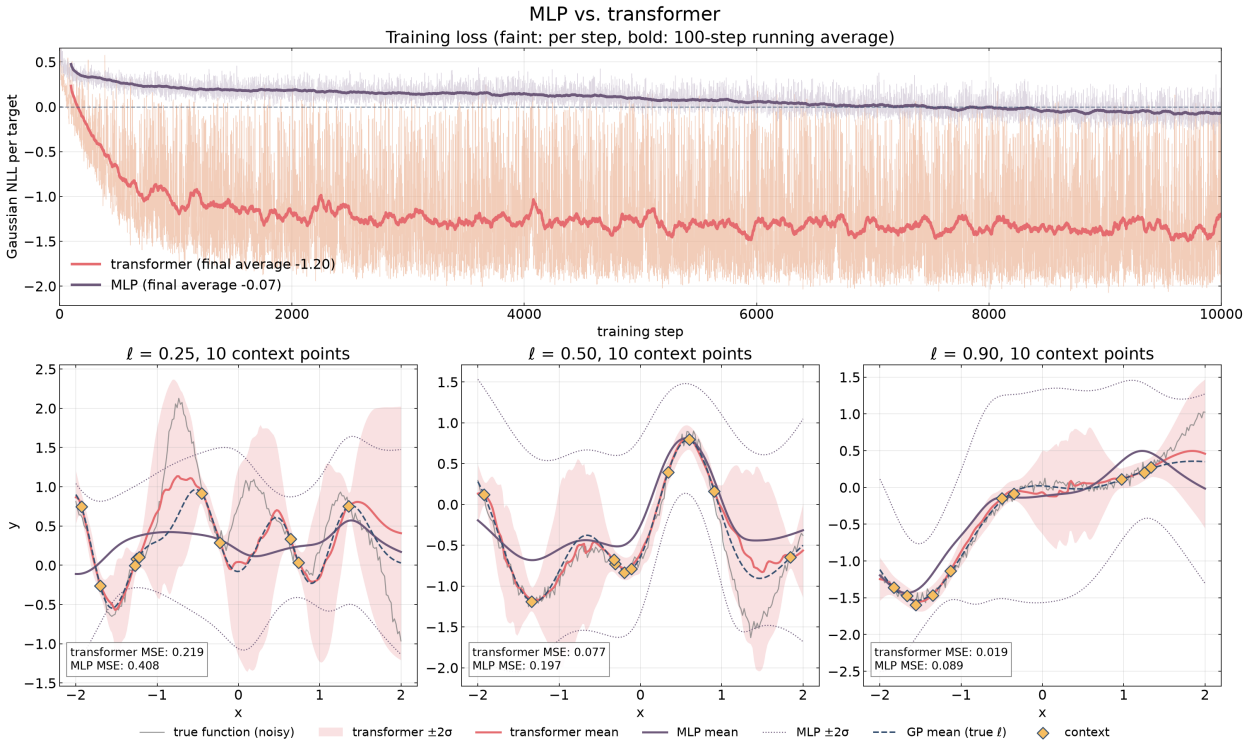


Figure 5: MLP vs. transformer: training loss of both models (top), and both models' predictions with $\pm 2\sigma$ on three functions with short, medium and long lengthscales (bottom).

Results

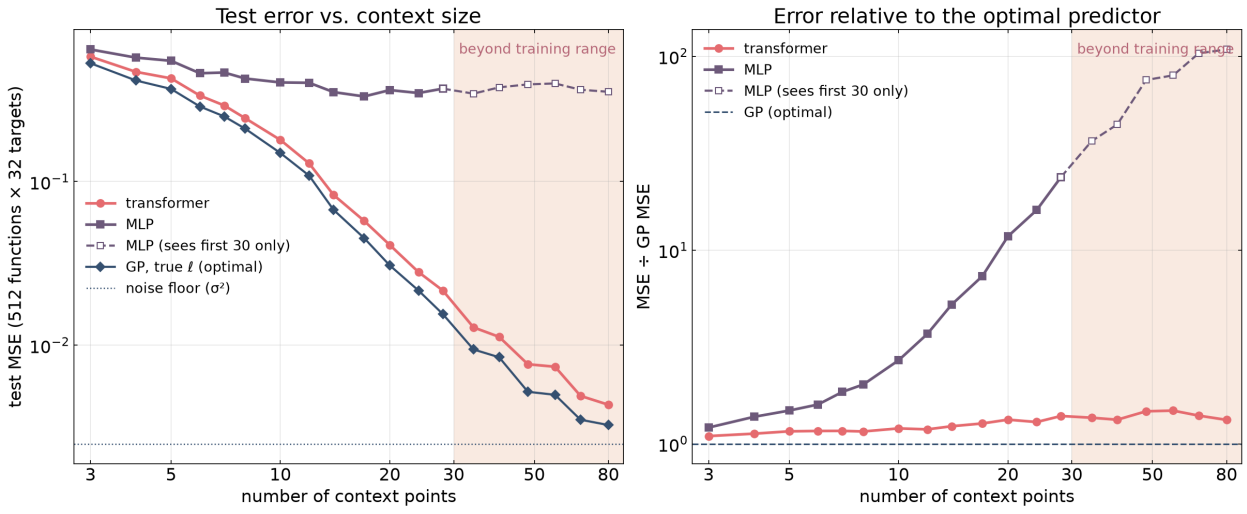


Figure 6: Test error against the number of context points (3 to 80) for the transformer, the MLP and the GP with the true lengthscale (left), and each model's error relative to the GP (right).

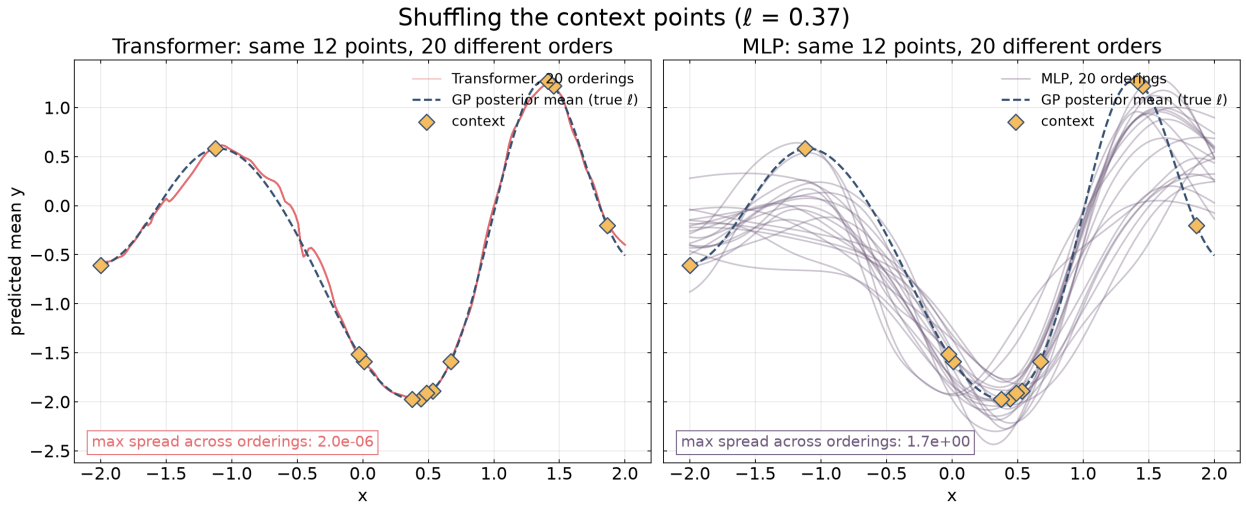


Figure 7: Predictions for one function with the same context points fed in 20 different orders: the transformer (left) and the MLP (right).

Additional plots

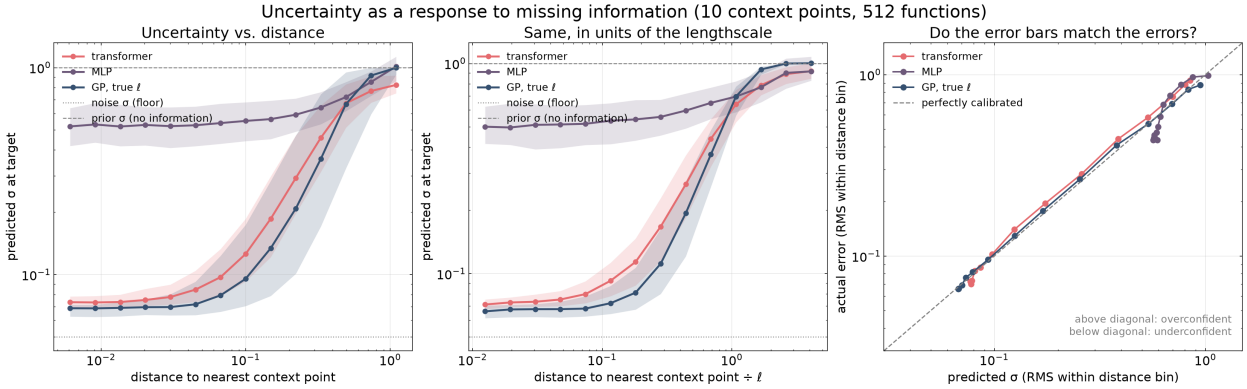


Figure 8: Predicted uncertainty against distance to the nearest context point, in raw units (left) and in units of ℓ (middle), and predicted σ against actual error (right).

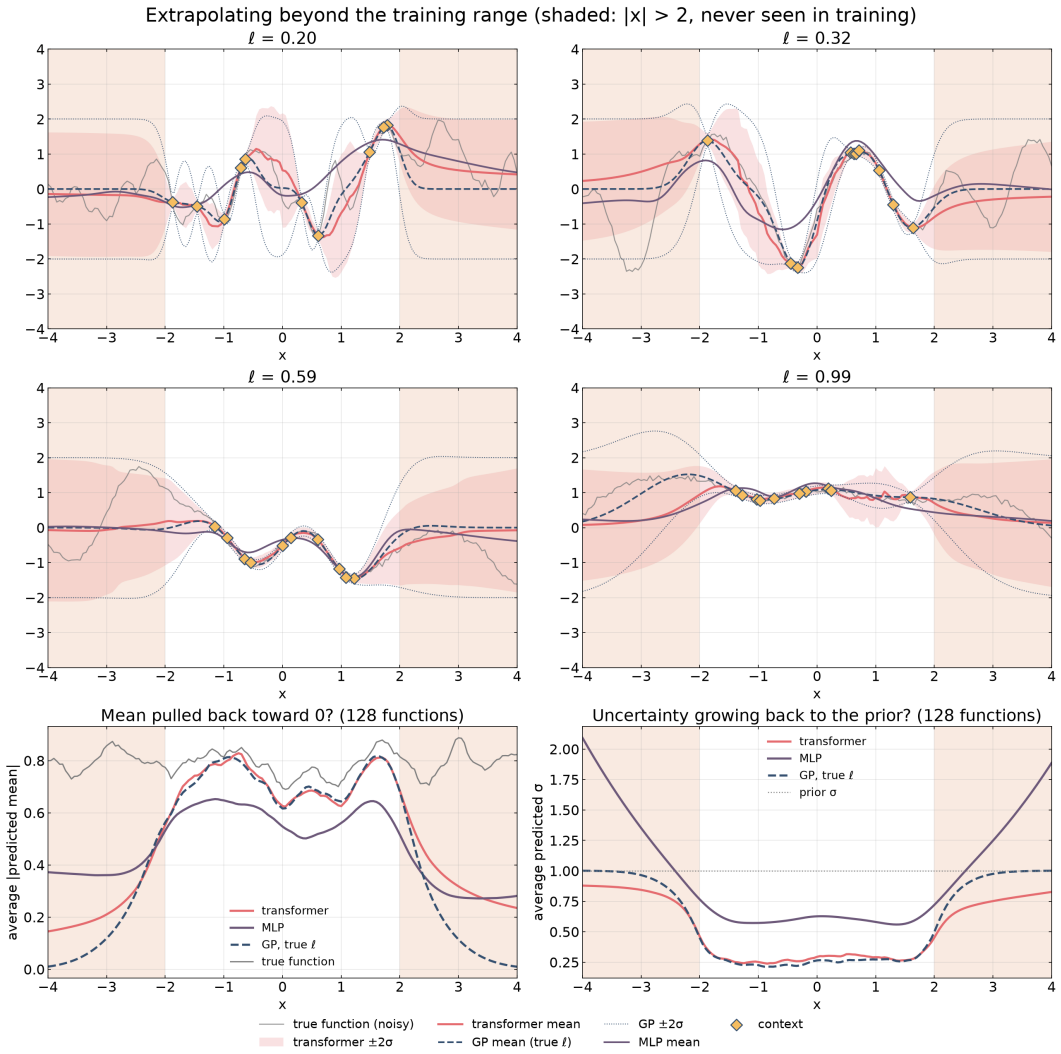


Figure 9: Predictions extrapolated to $x \in [-4, 4]$, beyond the training range $[-2, 2]$: four example functions, and the average |mean| and σ over 128 functions.